Introduction to Projects & Project Management Concepts

SECTION: What is a Project?

A project is a temporary endeavor undertaken to create a unique product, service, or result. Two defining characteristics separate a project from routine operations:

- **Temporary**: every project has a definite start and end. The end is reached when objectives are met, when it becomes clear objectives cannot be met, or when the need for the project no longer exists.

- **Unique**: the deliverable differs in some distinguishing way from other similar products or services, even if the category (e.g. "build a house") is common.

Projects are distinct from **operations**, which are ongoing and repetitive (e.g. manufacturing the same product daily). Operations keep an organization running; projects change it.

SECTION: The Triple Constraint (Iron Triangle)

Every project is bound by three interrelated constraints:

1. **Scope** — what work must be done.

2. **Time** — the schedule for delivering the work.

3. **Cost** — the budget available.

A fourth dimension, **Quality**, sits at the centre of the triangle: changing any one of scope, time, or cost affects quality and the other two constraints. For example, compressing the schedule without adding budget usually forces scope cuts or quality risk.

SECTION: Project Life Cycle

Most projects pass through four generic phases:

1. **Initiating** — defining the project at a high level, producing a project charter, identifying stakeholders.

2. **Planning** — developing the project management plan: scope, schedule, cost, risk, quality, communication, procurement, and HR plans.

3. **Executing** — carrying out the plan, coordinating people and resources.

4. **Monitoring & Controlling** — tracking, reviewing, and regulating progress; managing changes.

5. **Closing** — formally finalizing all activities, obtaining acceptance, releasing resources.

(Monitoring & Controlling overlaps with Executing throughout the project rather than being strictly sequential.)

SECTION: Key Roles

- **Project Sponsor**: provides resources and authority, championing the project at the executive level.

- **Project Manager (PM)**: responsible for meeting project objectives; manages the triple constraint, the team, and stakeholder expectations.

- **Stakeholders**: any individual or group affected by or able to affect the project (customers, team members, suppliers, regulators, the public).

- **Project Management Office (PMO)**: an organizational unit that standardizes project governance, provides templates, and sometimes directly manages projects.

SECTION: Organizational Structures and PM Authority

The PM's authority varies depending on the organizational structure:

- **Functional**: staff report to functional managers; PM has little or no formal authority (a coordinator role).

- **Matrix (weak/balanced/strong)**: authority is shared between functional managers and the PM; a "strong matrix" gives the PM more power.

- **Projectized**: the PM has full authority over the project team, who report directly to the PM for the project's duration.

SECTION: Success Criteria

A project's success is no longer judged only by whether it finished on time and on budget (the traditional Iron Triangle view). Modern definitions also weigh:

- Stakeholder satisfaction

- Business value delivered (did it achieve the intended benefit?)

- Quality of the deliverable

- Team learning and organizational capability gained

SECTION: Common Exam Angles

- Distinguish "project" vs "program" (a program is a group of related projects managed in a coordinated way for benefits not available from managing them individually) vs "portfolio" (a collection of projects/programs grouped to achieve strategic objectives, not necessarily interdependent).

- Explain why the triple constraint is a trade-off, not three independent variables.

- Map a described scenario to the correct life-cycle phase.